

Q1. Constrained Optimisation Basics. Consider the following optimization problem:

$$\min_{x_1, x_2} x_2^2 + x_1$$

subject to

$$x_1 \geq 1$$

$$x_1 x_2 \geq 1$$

- a) Write the Lagrangian function of this problem.
- b) Find the solution x_1, x_2 analytically.
Note: this is **harder** than any test question.
- c) What type of problem is this?
- d) Suggest an algorithm to solve problems of this type. Explain i) the key concepts of your chosen algorithm, ii) why it may be a good choice to solve problems of this type.

Q2. Duality. Consider the same problem as in Q1, except with the second constraint removed.

- a) Is the problem convex? If so, state how you know this. If not, state why.
- b) Calculate the Lagrangian dual function of this problem.
- c) Calculate, or otherwise determine, the duality gap.

Q3. Optimal control.

- a) Explain the principles of model predictive control, including a mathematical formulation. List at least two main advantages and at least two disadvantages.
- b) Consider the system $x_{k+1} = x_k + u_k + 1$ where x_k, u_k are restricted to integers. The cost function is $c(x, u) = x^2 + u^2$ and constraints as follows:

$$0 \leq x_k \leq 2$$

$$-2 \leq u_k \leq 1$$

If the terminal cost is $2x$, and the prediction horizon is 2, work out the MPC control law assuming that the initial state is within $0 \leq x_k \leq 2$.

- c) (Advanced): Usually in practice a terminal constraint or cost is added, i.e. at the final state of the horizon there is a penalty and/or constraint to keep the final state within a desired region. Using your understanding of how MPC works, why might this be a good idea? Can you think of any disadvantages?
- d) Explain Bellmann's principle of optimality in your own words.

Q4. Reinforcement Learning

- a) When would you use value iteration or policy iteration, as opposed to e.g. Q learning?
- b) Explain how policy iteration works, giving an algorithm (either conceptually or mathematically).
- c) In Q Learning, why is it generally better to use an ϵ -greedy policy rather than a 100% greedy policy?
- d) (Advanced): Q Learning is an off-policy method. Adapt Q Learning to use its own policy and thus become an on-policy method. Then, look up the SARSA algorithm and compare.